

Polar Doctor

Installation and user manual

Sailing-boat performance polars,
built from your own sailing data

Version 2.1.2 — September 2026
https://github.com/ozolli/polar_doctor

MIT

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1. Overview

Polar Doctor builds, edits and analyses the **performance polars** of a sailing boat from your own sailing data: NMEA 0183 log files, VDR databases recorded by qtVlm, NMEA 2000 streams, or live capture while you are sailing.

Since version 2.0 it is a **web application**. A small server, `polar_doctor_web`, runs on the onboard computer; the interface opens in any browser, at the chart table or on a tablet in the cockpit. Nothing needs to be installed on the client devices.

1.1 What is a polar?

A boat's polar is a table — and a diagram — giving the **boat speed** for every combination of wind **angle** and wind **strength**. It is the performance identity card of the boat. Routing software uses it to compute a route; on board, it tells you whether you are sailing to the boat's potential, and lets you compare trim settings.

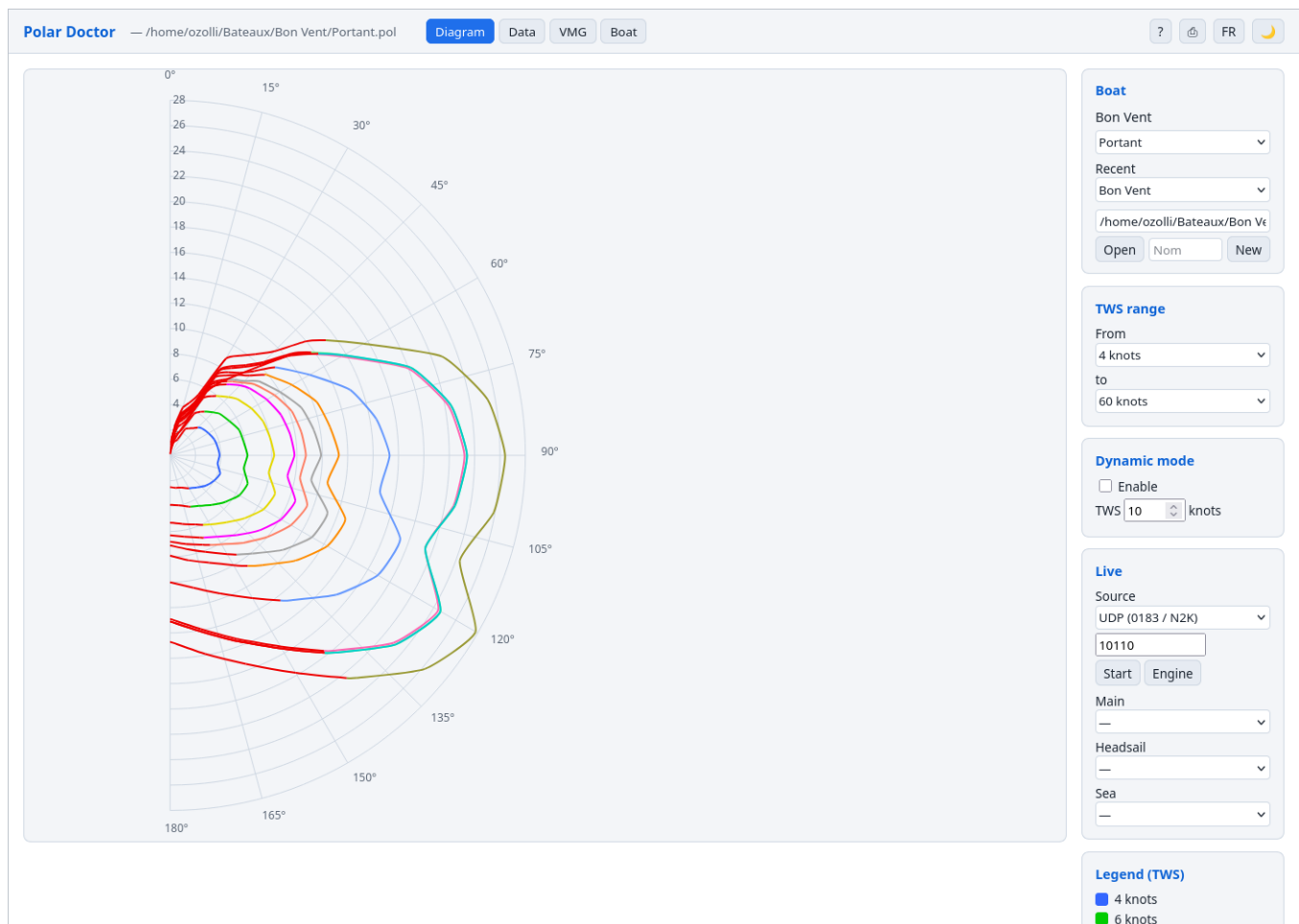
A polar supplied by a builder is theoretical: it describes a new boat, light, sailed by a trained crew on flat water. The one Polar Doctor builds describes your boat as it actually sails, with its load, its sails and its crew.

1.2 Vocabulary

Term	Meaning
TWA	True Wind Angle — angle of the true wind to the boat's axis, from 0° (head to wind) to 180° (dead downwind).
TWS	True Wind Speed, in knots.
STW / BSP	Speed Through Water, measured by the log. This is what the polar describes.
SOG	Speed Over Ground, from the GPS. Used to sanity-check the STW.
VMG	Velocity Made Good — the part of the boat speed gained along the wind axis. Measures upwind and downwind efficiency.
AWA / AWS	Angle and speed of the apparent wind, the one felt on board.

1.3 What the program does, in short

1. It reads your data: log files, VDR database, or a live stream.
2. It keeps only the useful moments: engine excluded, log glitches rejected, measurements smoothed.
3. It sorts every measurement into a cell of 5° of angle by 2 knots of wind.
4. For each cell it keeps a **high percentile** of the observed speeds (P90 by default): the achievable performance, not the average.
5. It writes the result to a `.pol` file, readable by qtVlm and other routing software.



The main window: polar diagram, boat selector, wind range, Live card and legend.

2. Installation

Polar Doctor runs on **Linux** (x86-64 and ARM64, so also on a Raspberry Pi) and on **Windows** (64-bit). The server only needs two libraries, GLib and SQLite; it uses no graphical toolkit.

2.1 Ready-made binaries

The project's Releases page (https://github.com/ozolli/polar_doctor/releases) provides:

- `polar_doctor-linux-x64.tar.gz` — 64-bit Linux;
- `polar_doctor-linux-arm64.tar.gz` — Linux ARM64 (Raspberry Pi);
- `polar_doctor-windows-x64.zip` — 64-bit Windows, executable and libraries, nothing to install.

Every archive comes with a `.sha256` file so you can check it:

```
sha256sum -c polar_doctor-linux-x64.tar.gz.sha256
```

Linux

```
tar -xzf polar_doctor-linux-x64.tar.gz
cd polar_doctor-linux-x64
./polar_doctor_web ~/Boats/MyBoat
```

Then open `http://localhost:8081` in a browser.

Windows

1. Unzip `polar_doctor-windows-x64.zip` into a folder of your choice.
2. Double-click `polar_doctor_web.exe`. A console window opens and prints the listening address.
3. Open `http://localhost:8081`.

To open a boat at startup, pass the folder as an argument, for instance from the command prompt:

```
polar_doctor_web.exe C:\Boats\MyBoat
```

2.2 Building from source

Linux (Debian, Ubuntu, Raspberry Pi OS)

```
sudo apt install build-essential pkg-config libglib2.0-dev libsqlite3-dev
git clone https://github.com/ozolli/polar_doctor.git
cd polar_doctor
make
./polar_doctor_web ~/Boats/MyBoat
```

Windows (MSYS2)

Install MSYS2, then in a MINGW64 shell:

```
pacman -S --needed make mingw-w64-x86_64-gcc mingw-w64-x86_64-pkgconf \
mingw-w64-x86_64-glib2 mingw-w64-x86_64-sqlite3
make
```

To ship the executable outside MSYS2, copy the libraries it uses next to it:

```
ldd polar_doctor_web.exe | grep -i /mingw64/ | awk '{print $3}' | xargs -I {} cp {} dist/
```

Tests

`make test` builds and runs the core unit tests (NMEA 2000 decoding, Actisense protocol). They must all print OK.

2.3 Starting automatically (systemd service, Linux)

So that Polar Doctor starts with the onboard computer and stays reachable from the network:

```
make                # as a normal user, never under sudo
sudo make install    # installs the binary and the service for that user
sudo systemctl enable --now polar_doctor_web
```

`sudo make install:`

- copies the binary into `/usr/local/bin/`;
- installs `/etc/systemd/system/polar_doctor_web.service`, set to run under **your** account, not root;
- creates `/etc/default/polar_doctor_web` (mode 600) if it does not exist;
- never overwrites an existing `/etc/default/polar_doctor_web`: your settings and your password survive upgrades.

The service listens on all interfaces, port 8081, and reopens the most recently used boat. Useful commands:

```
sudo systemctl status polar_doctor_web    # state
sudo systemctl restart polar_doctor_web    # after an upgrade
journalctl -u polar_doctor_web -f         # live log
```

Upgrading. After a `git pull`: `make && sudo make install && sudo systemctl restart polar_doctor_web`. Your polars, configuration and password are left untouched.

3. Access, password and settings files

The interface **writes** files on the server (polars, boat configuration), creates folders and reads paths you give it. Exposing it to the network unprotected would hand those rights to the whole network.

- Locally (address `127.0.0.1`, the default), no password is asked.
- To listen on the network a password is **mandatory**: without one the server refuses to start, unless explicitly given `--allow-anonymous`.

3.1 Where to put the password

It is always written as `WEB_AUTH=user:password`, and never given on the command line (which other users of the machine can read).

Linux, as a service

```
sudo nano /etc/default/polar_doctor_web    # WEB_AUTH=me:MyPassword
sudo systemctl restart polar_doctor_web
```

Linux, started by hand

```
read -rsp 'Password: ' P; echo
WEB_AUTH="me:$P" ./polar_doctor_web ~/Boats/MyBoat --bind 0.0.0.0
```

Windows

On its first run the server creates a fully commented settings file and prints its path. Open it (Windows key + R):

```
notepad %LOCALAPPDATA%\polar_doctor\web.conf
```

then remove the `#` in front of the lines you need:

```
WEB_AUTH=me:MyPassword
BIND=0.0.0.0
```

Restart `polar_doctor_web.exe`. On the first network access, Windows Firewall asks you to allow the program.

3.2 The `web.conf` file

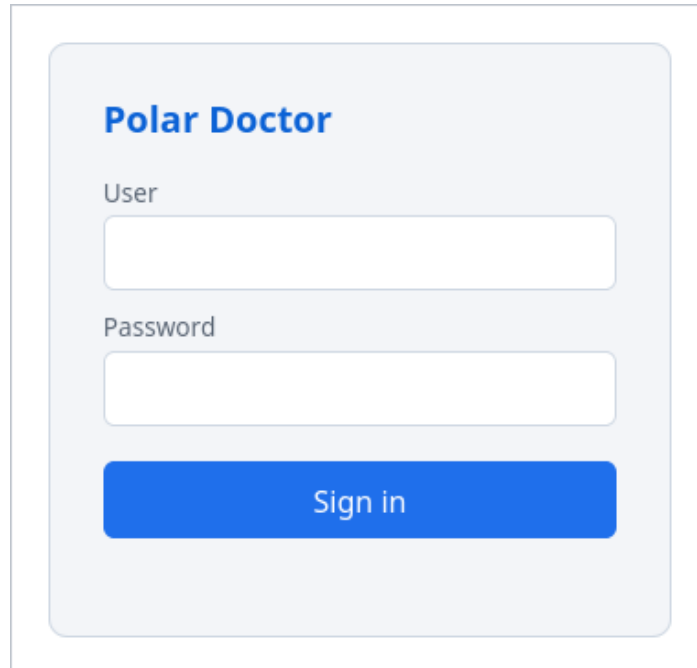
Same keys on both systems. The file lives in `%LOCALAPPDATA%\polar_doctor\web.conf` on Windows and `~/.config/polar_doctor/web.conf` on Linux; `--config FILE` selects another one.

Key	Role	Default
<code>WEB_AUTH</code>	<code>user:password</code> , mandatory outside <code>127.0.0.1</code>	—
<code>BIND</code>	listening address; <code>0.0.0.0</code> = the whole network	<code>127.0.0.1</code>
<code>PORT</code>	listening port	<code>8081</code>
<code>POLAR_DOCTOR_BOAT</code>	boat opened at startup	most recent

Precedence, strongest first: **command line**, then **environment variables**, then **web.conf**. On Linux a `web.conf` holding a password must be readable by its owner only (`chmod 600`), otherwise the server refuses to start.

3.3 Signing in

When a password is configured, the browser shows a sign-in page.

The image shows a web form for signing in to Polar Doctor. It has a light blue header with the text "Polar Doctor". Below the header, there are two input fields: "User" and "Password". The "User" field is a simple text box, and the "Password" field is a text box with a small eye icon on the right side to toggle visibility. Below these fields is a large blue button with the text "Sign in" in white.

The sign-in page. A cookie keeps the session open for a year on that device.

After signing in, a cookie is stored on the device (valid one year, not readable by scripts, not sent to other sites), so the password is not asked again on every visit. Changing the password signs every device out. The sign-out button is at the top right of the interface.

For scripts, plain HTTP authentication still works: `curl -u me:MyPassword http://boat:8081/api/polar`.

3.4 All of Polar Doctor's files

File	Location	Role
<code>web.conf</code>	Linux: <code>~/.config/polar_doctor/</code> Windows: <code>%LOCALAPPDATA%\polar_doctor\</code>	server settings
<code>polar_doctor_web</code>	<code>/etc/default/</code> (Linux)	systemd service settings, mode 600
<code>web_secret</code>	next to <code>web.conf</code>	key for the sign-in cookies. Deleting it signs every device out
<code>recent_boats</code>	next to <code>web.conf</code>	the last eight boats opened
<code>boat.cfg</code>	boat folder	inventory and polar definitions
<code>*.pol</code>	boat folder	the polars themselves

4. Getting started

For Polar Doctor a **boat** is simply a **folder** holding a `boat.cfg` configuration file and one or more `.pol` polars. Nothing else is needed, and the folder can be copied or backed up like any other.

4.1 Creating a boat

In the **Boat** card, top right:

1. type the path of the folder to create — for instance `/home/sailor/Boats/Fair Wind` or `C:\Boats\Fair Wind`;
2. type the boat's name in the Name field;
3. click **New**.

The folder is created, with a default `boat.cfg` and an empty polar, and the boat is opened straight away.

4.2 Opening an existing boat

Type its folder and click **Open**, or pick it from the **Recent** list, which keeps the last eight. The boat you open is also the one the server will reopen next time it starts — handy for a service that restarts with the boat.

A single `.pol` file can also be opened without a boat folder: `./polar_doctor_web polar.pol`. Boat-related features (inventory, multiple polars, live routing) are then unavailable.

4.3 The screen at a glance

- Top: the name of the open file, the four tabs (**Diagram**, **Data**, **VMG**, **Boat**), and on the right the **?** (help), print, **FR/EN** and light/dark buttons.
- Left: the main view, depending on the selected tab.
- Right: the **Boat**, **TWS range**, **Dynamic mode**, **Live** and **Legend** cards.

Language and theme are remembered by each browser separately: the cockpit tablet can be in dark theme while the chart-table PC stays light.

5. Describing the boat

The **Boat** tab describes the sail inventory and the polars you want to keep. This is what lets the program file every measurement into the right polar during a live capture.

Polar Doctor — /home/ozolli/Bateaux/Bon Vent/Portant.pol Diagram Data VMG **Boat** ? ⓘ FR 🌙

Save + Polar

Name Bon Vent

Main

GV
1R
2R
3R

Headsail

J1
J3
Code0
Spi

Sea

Calme
Belle
Peu agitée
Agitée

Engine

Engine Moteur

Charge Charge

Polars

Name Pres

X

Main

☒ GV ☒ 1R ☒ 2R ☒ 3R

Headsail

☒ J1 ☒ J3 ☐ Code0 ☐ Spi

Sea

☐ Calme ☐ Belle ☐ Peu agitée ☐ Agitée

The Boat tab: name, inventory (mainsail, headsails, sea states), engine keywords and polars.

5.1 Inventory

- **Main:** mainsail states, one per line — for instance `Full`, `R1`, `R2`, `R3`.
- **Headsail:** the headsails — `J1`, `J3`, `Code0`, `Spi`...
- **Sea:** sea states. The Douglas scale is used by default.
- **Engine:** the two keywords looked for in the comments of a VDR database. One means the propeller is engaged (those points are dropped), the other means the engine is running in neutral to charge the batteries (those points are kept, because the boat is still sailing).

5.2 Polars and criteria

Each polar has a name — which is also its file name, `Upwind.pol`, `Downwind.pol` — and **criteria** as checkboxes: which mainsails, which headsails and which sea states it covers.

Checking nothing means "everything". A polar with no criterion ticked accepts every condition. That is the right setting if you keep a single polar for the boat.

A criterion written in `boat.cfg` but missing from the inventory is shown in orange: either add it to the inventory, or untick it to remove it.

Save rewrites `boat.cfg`. The file is read back and checked first: if it is inconsistent, the old one is kept. Comments you may have added by hand to `boat.cfg` are lost when saving from the interface.

6. Building a polar from files

Data tab, **Import files** box. List the files to read, **one path per line**. These are paths on the server, not on the device showing the interface: nothing is uploaded.

The screenshot shows the Polar Doctor web interface. At the top, there's a navigation bar with tabs: Diagram, Data (selected), VMG, and Boat. Below the navigation bar, there's a header area with 'Save', '+ TWA', and '+ TWS' buttons. The main content area is titled 'Import files' and contains a text input field with the placeholder '/chemin/nav.nmea (un par ligne)'. Below the input field are 'Create' and 'Update' buttons, and a 'Percentile' dropdown menu set to 'P90'. Below the import section is a large table representing a polar plot. The table has 14 columns for wind speeds (4, 6, 8, 10, 12, 14, 16, 20, 26, 30, 36, 40, 60) and 14 rows for headings (0°, 5°, 10°, 15°, 20°, 25°, 30°, 35°, 40°, 45°, 50°, 60°, 70°, 80°, 90°, 100°, 110°). Each cell in the table contains a numerical value representing speed over ground.

	4	6	8	10	12	14	16	20	26	30	36	40	60
0°	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5°	0.40	0.60	0.80	0.90	1.00	1.00	1.10	1.10	1.20	1.30	1.20	0.10	0.00
10°	0.70	1.10	1.50	1.80	1.90	1.90	2.10	2.10	2.20	2.60	2.30	0.50	0.00
15°	1.10	1.70	2.30	2.70	2.80	2.90	3.10	3.30	3.40	3.90	3.40	1.20	0.00
20°	1.20	1.90	2.60	3.00	3.20	3.30	3.60	3.70	3.80	4.40	3.90	1.80	0.00
25°	1.40	2.30	3.10	3.70	3.80	4.00	4.30	4.40	4.60	5.30	4.70	2.70	0.00
30°	2.40	3.80	5.20	6.10	6.40	6.60	7.10	7.40	7.70	8.80	7.80	5.80	0.00
35°	2.60	4.20	5.70	6.80	7.10	7.30	7.70	8.00	8.40	9.60	8.60	7.10	0.00
40°	2.80	4.50	6.10	7.30	7.60	7.80	8.30	8.50	9.10	10.40	9.30	8.30	0.00
45°	3.10	4.80	6.50	7.80	8.10	8.30	8.90	9.40	10.20	11.60	10.40	9.90	0.00
50°	3.30	5.10	6.90	8.20	8.70	9.10	9.90	10.80	12.30	14.00	12.50	12.20	0.00
60°	3.50	5.40	7.40	8.80	9.50	10.00	11.10	12.70	15.60	17.60	15.80	15.80	0.00
70°	3.70	5.80	7.80	9.30	10.20	11.00	12.30	15.10	20.00	22.80	20.20	20.20	0.00
80°	3.80	5.90	8.00	9.60	10.50	11.50	12.80	16.50	22.30	25.40	22.50	22.50	0.00
90°	3.90	6.10	8.20	9.80	10.70	11.90	13.30	17.30	23.20	26.40	23.40	23.40	0.00
100°	3.80	5.90	7.90	9.40	10.40	11.30	12.60	16.80	22.70	25.90	22.80	22.80	0.00
110°	3.80	5.90	7.90	9.40	10.40	11.30	12.60	16.80	22.70	25.90	22.80	22.80	0.00

The Data tab: file import, percentile selector, and the editable polar table.

6.1 Accepted formats

- **NMEA 0183** (`.nmea`, `.log`, `.txt`): MWV or MWD wind, VHW speed through water, HDT / HDG heading, and speed over ground when present.
- **NMEA 2000** as **YDRAW** text, as recorded by Yacht Devices gateways or published by n2k-mux (port 2700). One frame per line.
- **qtVlm VDR** (`.db`): SQLite databases recorded while sailing, with wind, speed, and possibly engine revolutions and comments.

What about Actisense gateways? An **NGX-1** or **NGT-1** is read **live**, over the serial port (see chapter 7). Its binary logs, however — Actisense `.ebl` files, or a raw capture of the serial port — cannot be imported here: file import expects text. To record a passage for later import, write NMEA 0183 or YDRAW, or let qtVlm keep its VDR.

6.2 Create or update

- **Create** builds a brand-new polar from the listed files only. It is saved under a timestamped name in the boat folder.

- **Update** starts from the currently selected polar, adds the new measurements and recomputes everything. The polar can therefore go **up or down**: this is what gradually brings a theoretical polar back to the reality of the boat.

6.3 The percentile

Each cell of the table (5° of angle × 2 knots of wind) collects many measurements. To keep a single figure, Polar Doctor takes a **high percentile**, adjustable from P85 to P95, P90 by default.

- The average would be dragged down by manoeuvres, mishandled gusts or a hesitant autopilot.
- The maximum would keep a surfing gust or a measurement error.
- P90 means "the speed the boat exceeds one time in ten in these conditions" — the achievable performance when everything is properly trimmed.

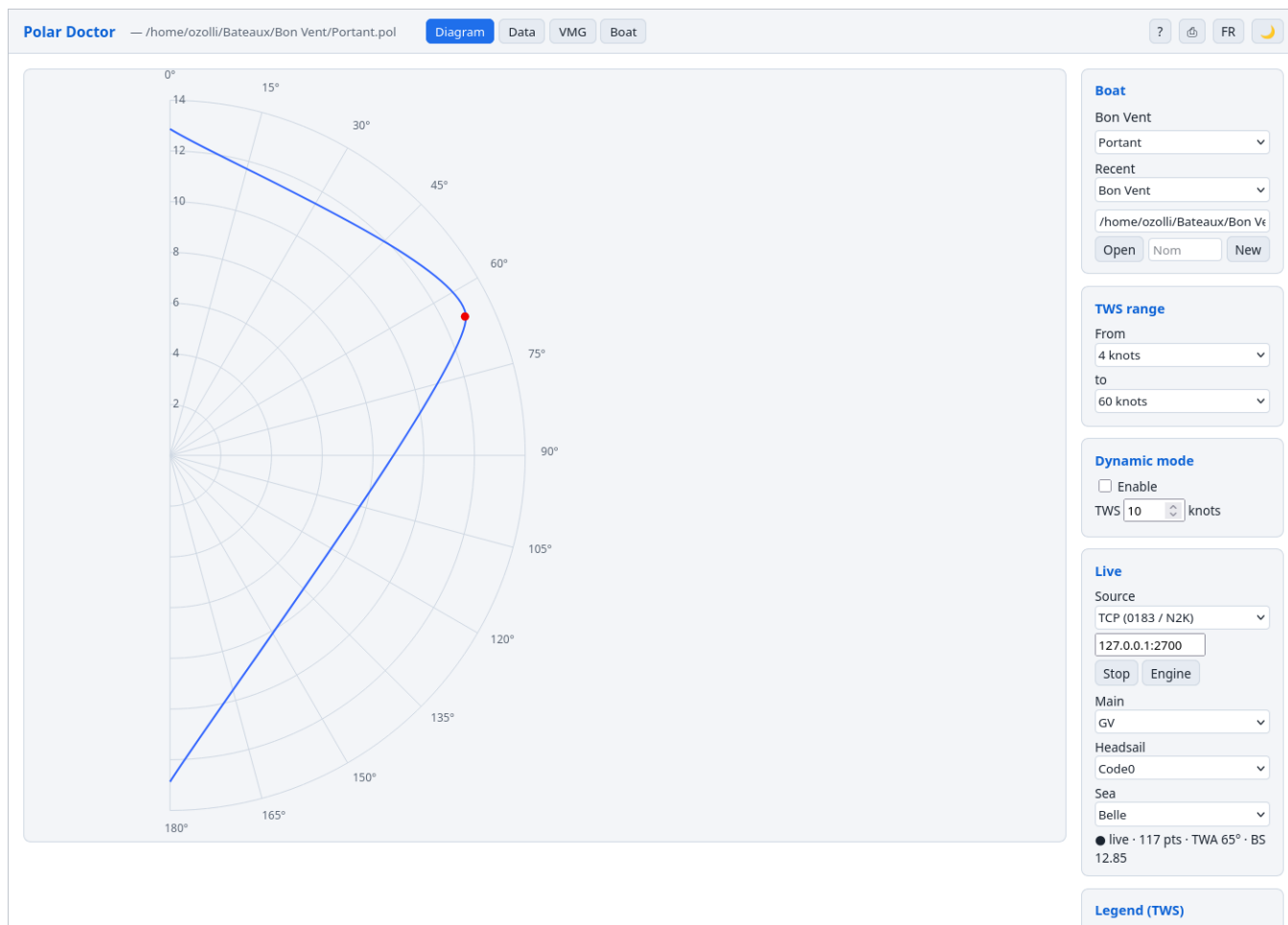
Lowering to P85 gives a cautious polar, useful for shorthanded offshore routing. Raising to P95 gives a racing polar, demanding to sail to. A cell holding fewer than **three** measurements is left empty: a single point is not a performance.

6.4 What is discarded automatically

- **Engine**: in a VDR database, any point recorded with non-zero engine revolutions is excluded, unless the comment says the batteries are being charged.
- **Log glitches**: the gap between speed through water and speed over ground is tracked continuously; a gap that changes abruptly (blocked paddle wheel, planing, air bubble) gets the measurement rejected. A current, which varies slowly, is kept.
- **Noise**: speed through water from NMEA files is smoothed by a moving average, reset at each manoeuvre so as not to smooth across a tack.
- **Impossible values**: angle outside 0–180°, wind above 70 knots, speed above 50 knots, NMEA sentences with a bad checksum.

7. Live capture

The **Live** card builds the polar while you sail. Choose a source, an address, then click **Start**.



Capture running: scatter of points, current point in red, the polar being built, and the counter at the bottom of the Live card.

7.1 Sources

Source	Address to type	Use
UDP (0183 / N2K)	a port number, e.g. 10110	listens to frames broadcast on the boat network
TCP (0183 / N2K)	host:port, e.g. nav.local:2700	connects to a multiplexer or a gateway
qtVlm VDR	path to vdr.db	follows the file qtVlm is writing
Actisense NGX-1 (serial)	/dev/ttyUSB0, COM3, optionally followed by @230400	reads a USB NMEA 2000 gateway directly

Over UDP and TCP the format is detected line by line: NMEA 0183 and NMEA 2000 (YDRAW) may even be mixed in the same stream.

About NMEA 2000

Polar Doctor does not attach to the CAN bus itself. It needs either:

- a gateway publishing N2K on the network as YDRAW — n2k-mux (port 2700), a Yacht Devices Wi-Fi gateway...;
- an Actisense NGX-1 or NGT-1 serial gateway, connected by USB and set to **Transfer** mode (raw data), which Polar Doctor reads directly;
- or a gateway converting N2K to NMEA 0183 broadcast on the network.

A serial port can only be opened once. If another program (n2k-mux, qtVlm) already holds the USB gateway, Polar Doctor cannot open it: read the network stream that program publishes instead.

Serial port permissions on Linux

```
sudo usermod -aG dialout $USER      # then log in again, or restart the service
```

Prefer a stable port name such as `/dev/serial/by-id/usb-Actisense_NGX-1-...`: unlike `/dev/ttyUSB0`, it does not change between plug-ins.

7.2 Boat state

Below the Live buttons, three lists give the **mainsail**, the **headsail** and the **sea state**. Every measurement is filed into **all** the polars whose criteria match that state. Change those lists at each manoeuvre: the capture carries on and the following points go to the right polar.

If the counter stays at 0, the chosen state usually matches no polar — or only polars other than the one displayed. The Live card then says so explicitly. Remember that the counter and the scatter only show the **displayed** polar.

7.3 The Engine button

When you engage the propeller, click **Engine**: capture carries on but points are no longer recorded. Click again when you stop. Without this, a few hours of motoring are enough to inflate the light-wind part of the polar.

7.4 Stopping and saving

Stop aggregates the measurements and writes the files:

- with a boat and its polars: every polar that received points is updated in place, starting from its existing content;
- without a boat: a timestamped `live_YYYYMMDD_HHMMSS.pol` file is created.

The names of the files written appear under the Live card.

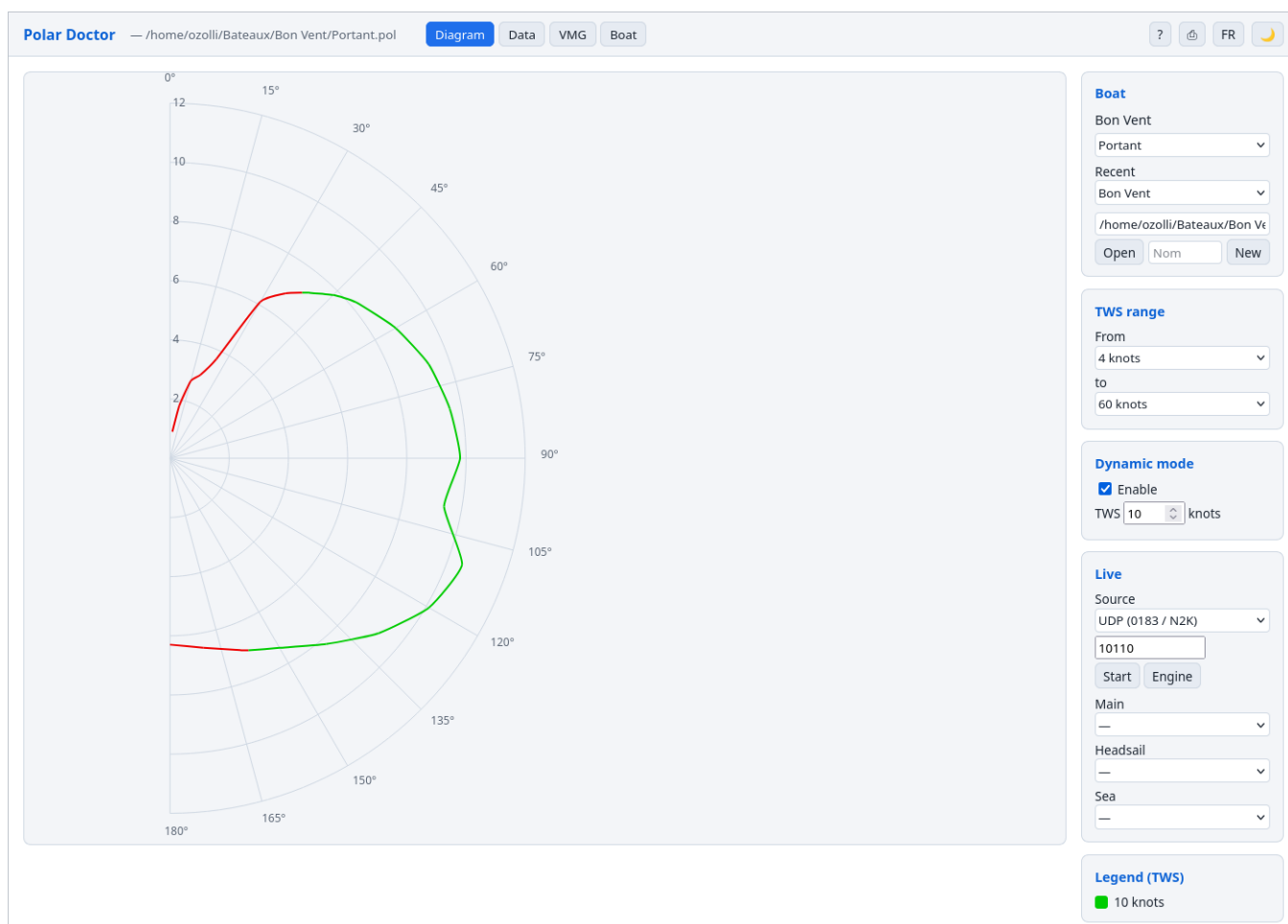
8. Reading the diagram

The **Diagram** tab draws the polar as a half rose: the bow points up (0°), dead downwind is at the bottom (180°), and the concentric circles give boat speed in knots.

- One **curve per wind strength**, with its colour, repeated in the legend.
- Curves are smoothed; the computed points are the angles present in the table.
- Stretches drawn in **red** mark the zones where VMG degrades: pinching further upwind, or bearing away further downwind, loses ground even if boat speed rises.
- The **TWS range** card restricts the display to a band of wind strengths, so the diagram is not buried under twenty curves.

8.1 Dynamic mode

Tick **Enable** in the Dynamic mode card and enter a wind strength, even one absent from the table: the matching curve is interpolated.



Dynamic mode: a single curve, interpolated for the requested wind strength.

Moving the mouse across the diagram, a line follows the aimed angle and shows the true wind angle, the apparent wind angle and speed, the boat speed and the VMG. This is the quick way to answer "at this angle, in this wind, what should I be doing?".

8.2 VMG table

The **VMG** tab gives, for each wind strength, the best upwind angle and the best downwind angle, with the matching boat speed.

Polar Doctor — /home/ozolli/Bateaux/Bon Vent/Portant.pol

Diagram Data **VMG** Boat

? ⓘ FR 🌙

TWS	Upwind °	BS	VMG	Downwind °	BS	VMG
4	45	3.10	2.19	150	2.99	2.59
6	38	4.39	3.45	158	4.35	4.04
8	38	5.92	4.68	155	6.07	5.49
10	39	7.16	5.60	158	7.01	6.49
12	38	7.38	5.83	156	7.74	7.07
14	37	7.53	5.99	146	9.42	7.83
16	50	9.80	6.36	145	10.65	8.72
20	50	10.80	6.94	143	14.37	11.45
26	54	13.65	8.00	142	19.61	15.41
30	54	15.26	9.08	142	22.40	17.55
36	54	13.79	8.13	142	19.76	15.57
40	56	14.18	8.03	142	19.76	15.57
60	0	0.00	0.00	180	0.00	0.00

Optimal upwind and downwind angles, computed for each wind strength.

These are the angles to aim for to gain to windward or to leeward as fast as possible. In practice: upwind, point up to the angle shown; downwind, bear away down to it. Those two columns are often the most useful output of the whole program.

9. Editing and saving the data

The **Data** tab shows the polar as an editable table: one row per angle, one column per wind strength. Click a cell to correct a value.

- **+ TWA** adds an angle, **+ TWS** a wind strength.
- The cross at the head of a row or column deletes that angle or strength.
- **Save** writes the `.pol` file.

Saving is protected: the file is written alongside first, read back and checked, then moved into place in one step. A power cut or an inconsistent polar cannot destroy the existing file.

When should you edit by hand? Usually to fill a hole: an empty cell for an angle or a wind strength you have never met, when the routing software needs a value anyway. Copying the neighbouring value is better than a zero, which would tell the router that the boat stops.

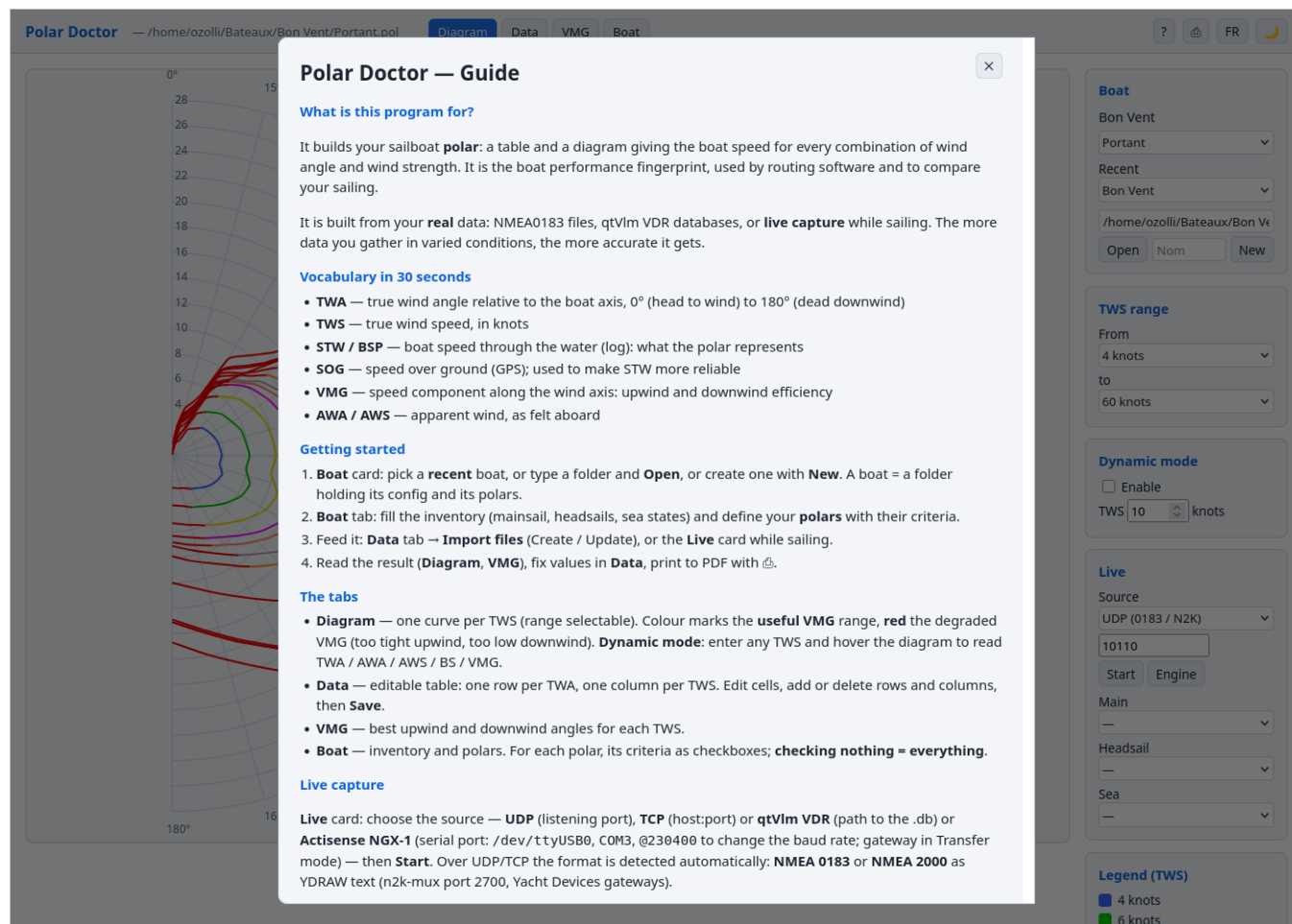
10. Printing and exporting to PDF

The print button at the top right starts the browser's print dialogue. Choose "Save as PDF" as the printer to get a file. Only the displayed view is printed — diagram, table or VMG table — without the side cards.

Handy to pin the VMG table at the chart table, or to carry a printed diagram in case of an electronics failure.

11. Help, language and theme

The **?** button opens the built-in help, which covers the essentials of this manual in a few screens.



The built-in help, available without an internet connection, in French and English.

The **FR/EN** button switches the language of the interface and of the help. The next button toggles light and dark themes — the dark theme preserves night vision at the chart table. Both settings are remembered by the browser.

12. File formats

12.1 Polar files (.pol)

A semicolon-separated table, readable by qtVlm and most routing software. The first column after the header is a **TWS 0 sentinel column**, always present and never displayed:

```
TWA\TWS;0;6;8;10;12
45;0.00;4.80;5.90;6.70;7.10
60;0.00;5.20;6.40;7.30;7.80
90;0.00;5.60;6.90;7.80;8.40
```

12.2 NMEA 0183

Sentence	Role
MWV (reference T)	true wind referenced to the water — preferred. Apparent wind (reference R) is ignored
MWD + HDT/HDG	ground-referenced true wind plus heading: used only when MWV , T is absent
VHW	speed through water — required
RMC , VTG , VBW , RMA , OSD	speed over ground, read when present

Water-referenced wind is preferred because the polar describes the boat in the water: ground-referenced wind is skewed by the current. Angles are folded from 0–360° to 0–180°, so both tacks feed the same polar.

12.3 NMEA 2000 (YDRAW)

One CAN frame per line: timestamp, direction, CAN identifier, bytes.

```
09:30:59.677 R 09FD0205 FF 73 04 B8 BD FC FF FF
09:30:59.677 R 09F50305 FF A6 03 FF FF 00 FF FF
```

PGN	Role
130306 Wind Data	water-referenced true wind first; ground-referenced as a fallback; apparent ignored
128259 Speed	speed through water — required
129026 COG & SOG	speed over ground
127250 Vessel Heading	true heading, or magnetic corrected by the variation

12.4 qtVlm VDR databases

An SQLite database, table **VDR**, columns **TWA**, **TWS**, **STW**, and where present **SOG**, **RPM** and **COMMENT**. Annotating your passages in qtVlm (sail name, sea state, engine keyword) lets Polar Doctor split the measurements between your polars automatically.

12.5 boat.cfg

```
[boat]
name = Fair Wind

[mainsail]
Full
R1
R2

[headsails]
J1
J3
Code0

[seastates]
Calm
Smooth
Moderate

[engine]
moteur = Engine
charge = Charging

[polars]
Upwind = mains: Full, R1, R2 ; heads: J1, J3 ; seas: *
Downwind = mains: Full, R1 ; heads: Code0 ; seas: *
```

The file is written by the Boat tab but stays editable by hand. An asterisk means "everything". Without a `[seastates]` section, the Douglas scale is used.

13. Command line

```
polar_doctor_web [boat-folder | file.pol] [--port N] [--bind ADDRESS]
                 [--config FILE] [--allow-anonymous]
```

Option	Role
boat-folder	boat opened at startup. Otherwise the <code>POLAR_DOCTOR_BOAT</code> variable, otherwise the most recent boat
<code>--port N</code>	listening port (default 8081)
<code>--bind ADDRESS</code>	listening address (default <code>127.0.0.1</code> ; <code>0.0.0.0</code> = whole network)
<code>--config FILE</code>	settings file to use
<code>--allow-anonymous</code>	allows network listening without a password — not recommended

14. Troubleshooting

Symptom	Likely cause and cure
The live counter stays at 0	The state (main, headsail, sea) matches no polar, or not the displayed one. The Live card says so. Set the state, or untick every criterion of a polar so that it accepts everything.
Live capture will not start	The message under the Live card gives the cause: connection refused, missing serial port, unreadable VDR file.
Permission denied on the serial port	The server's user is not in the <code>dialout</code> group.
The serial port is busy	Another program already holds the gateway. Read its network stream instead of the port.
Capture running, but no points	The stream may carry no true wind, or no speed through water. Both are required. Check that the instrument publishes true wind and that the log works.
The server refuses to start on the network	No password is set: fill in <code>WEB_AUTH</code> .
Forgotten password	Change it in the settings file and restart. Every device will have to sign in again.
Sign every device out	Delete the <code>web_secret</code> file and restart the server.
Empty cells in the polar	Fewer than three measurements in the cell, or conditions never met. Sail more, or fill them in by hand in the Data tab.
The polar is far too optimistic	Motoring was counted: use the Engine button live, annotate your VDR files, or lower the percentile.
The page is unreachable from the network	The server listens on <code>127.0.0.1</code> : set <code>BIND=0.0.0.0</code> . Check the firewall too.

14.1 Building a good polar

- It takes time: a few hundred miles in varied conditions are worth more than a day of trials.
- Sail at every angle. A polar fed only downwind will stay empty upwind.
- Check the log calibration: the whole polar depends on it. Compare speed through water and speed over ground in calm water without current, on a there-and-back run.
- Keep several polars (per headsail, for instance) rather than a single one mixing a spinnaker and a storm jib.
- Update regularly: a dirty hull, a cruising load or a new set of sails all show up in the figures.